MILITARY SPECIFICATION

YOKE, TOWING AND LIFTING

FOR DRUMS, FABRIC, COLLAPSIBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.2 Specification part number. The specification part number in this yoke is M40628.

1.1 Scope. This specification covers a yoke for ground towing and airlifting of filled collapsible fabric drums.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8110

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:27Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-T-97	- Tape, Pressure-Sensitive Adhesive Filament Reinforced.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-I-6868	- Inspection Process, Magnetic Particle.

STANDARDS

MILITARY

MIL-STD-22	- Welded Joint Design.
MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-6866	- Inspection, Liquid Penetrant.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13216E7991	- Yoke, Towing and Lifting: for Drums, Fabric, Collapsible.

(Copies of specifications, standards, and drawings required by contractors in connection with specific functions should be obtained from the procuring activity or as directed by by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:27Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, 2501 North West Seventh Street, Miami, FL 33125.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J429 - Mechanical and Material Requirements for Externally Threaded Fasteners.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The towing and lifting yoke shall be in accordance with Top Assembly TA13216E7991 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on drawings to insure correct fit, assembly, and operation of the yoke. Any data (e.g., shop drawings, layouts, flow sheets,

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:27Z
Check the source to verify that this is the current version before use.

processing procedures, etc.) prepared by the contractor or obtained from a vendo to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of yokes from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish yokes conforming to this specification.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The yoke shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the yoke may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the yoke may be newly fabricated from recovered materials to the maximum extent practicable, provided the yoke produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the yoke.

3.4 Weight. The weight of the completely assembled yoke shall be not more than 75 pounds.

3.5 Proof loads. The completed yoke shall show no permanent deformation when under a tension load of not less than 9,000 pounds or a compression load of not less than 4,000 pounds as specified in figure 1.

3.6 Identification markings. The yoke shall be identified in accordance with MIL-STD-130. The markings shall be applied to the yoke as shown on the drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:27Z
Check the source to verify that this is the current version before use.

3.7 Treatment and painting. The portions of the yoke normally painted shall be cleaned, treated and painted in accordance with referenced drawings. Unless otherwise specified (see 6.2), the finish paint shall be lusterless.

3.8 Workmanship. The yoke shall be clean and shall be free from grease, grit, dirt, oil, and any other foreign materials that could harmfully affect the handling of, or could impair serviceability of, the yoke. Holes shall be accurately drilled.

3.8.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are necessarily required.

3.8.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the yoke to conform to the examination and test requirements specified in section 4. Joints will be in accordance with MIL-STD-22. Weld procedures shall be in accordance with AWS D.1.

3.8.3 Welders. Before assigning welder to manual work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code.

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Certification shall be made available for review by the contracting officer or designated representative. Contractors who make only horizontal welds need not qualify welders for "all position welding". The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

3.8.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts not less than grade 5 conforming

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:27Z
Check the source to verify that this is the current version before use.

to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than 2 full threads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article units shall be subjected to tests in 4.5.2.1 through 4.5.2.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:27Z
Check the source to verify that this is the current version before use.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. AQL shall be 1.0.

4.4.3 Tests. Each yoke shall be tested as specified in 4.5.2. Failure of either test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The yoke shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
104. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.3.3).
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the yoke (see 3.3.4).
106. Weight not as specified (see 3.4).
107. Components missing or not as specified.
108. Dimensions not as specified (see 3.1.1).
109. Identification marking missing, incomplete, or not legible (see 3.6).
110. Treatment and painting not as specified (see 3.7).
111. Workmanship not as specified (see 3.8).
112. Metal fabrication not as specified (see 3.8.1).
113. Welding not as specified (see 3.8.2).
114. Welder's certification not made available for review (see 3.8.3).
115. Bolted connections not as specified (see 3.8.4).
116. Welding not in accordance with welding codes (see 3.8.2).

4.5.2 Tests.

4.5.2.1 Tension proof load. Subject the completed yoke to a static tension load of 9,000 pounds, applied in the direction indicated in figure 1. Apply static tension load for not less than 15 seconds. The yoke shall be inspected by one of the following methods after the test:

a. Magnetic particle in accordance with MIL-STD-6868. Discontinuities will not be permitted.
b. One penetrant inspection in accordance with MIL-STD-6866, type I, method B. Discontinuities will not be permitted.

Nonconformance to 3.5 or either welding inspection shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:27Z
Check the source to verify that this is the current version before use.

4.5.2.2 Compression proof load. Subject the completed yoke to a compression load of 4,000 pounds, applied in the direction indicated in figure 1. Apply compression load for not less than 5 seconds. The yoke shall be inspected by one of the following methods after the test:

a. Magnetic particle in accordance with MIL-STD-6868. Discontinuities will not be permitted.
b. One penetrant inspection in accordance with MIL-STD-6866, type I, method B.
Discontinuities will not be permitted.

Nonconformance to 3.5 or either welding inspection shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 1.0.

117. Disassembly and securing not as specified for all degrees.
118. Unpainted metal surfaces requiring the application of a contact preservative in accordance with the referenced document not coated with preservative for level A.
119. Shipping containers not as specified for levels A or B.
120. Blocking and bracing not as specified for levels A or B.
121. Strapping not as specified for levels A or B.
122. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. Each yoke shall be disassembled by removing the pins from the end of the left and right brace; the yoke shall then be folded to its smallest dimensions. Removed pins shall be replaced in the appropriate holes and secured to prevent loss. The removed brace and folded yoke shall be secured together, in a manner to prevent unfolding, with soft annealed wire or with tape conforming to PPP-T-97, type IV, 1.50 inches wide.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:27Z
Check the source to verify that this is the current version before use.

5.1.1.2 Unpainted surfaces. Unpainted metal surfaces of the yoke requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in, and shall be applied in accordance with, MIL-P-116.

5.1.2 Commercial. Yokes shall be disassembled to their minimum dimension and secured as specified for level A. Preservation shall be in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The yokes, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-601, overseas type, sytle optional, or PPP-B-621, class 2, style optional. Gross weight of shipping containers shall not exceed 250 pounds. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B, size as applicable. Blocking and bracing of the contents shall be in accordance with MIL-STD-1186.

5.2.2 Level B. The yokes, preserved as specified in 5.1, shall be packed as specified in 5.2.1 for level A, except the boxes shall be domestic type, and strapping shall be finish A.

5.2.3 Commercial. Yokes shall be packed in containers with a gross weight not exceeding 250 pounds in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract orders, marking shall be in accordance with MIL-STD-129 for levels A and B. Marking of commercial packaging shall be in accordance with ASTM D 3951; in addition, shipping containers shall show the gross weight and cube.

6. NOTES

6.1 Intended use. The yoke is intended for ground towing and external helicopter airlifting of filled collapsible fabric drums.

6.2 Ordering data.

6.2.1 Aquisition requirements. Aquisition documents shall specify the following:

a. Title, number and date of this specification.
b. When a first article test is not required.
c. Finish paint if other than lusterless (see 3.7).
d. When the welding process will not be at the option of the contractor (see 3.8.2).
e. Degree of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of two units. The contracting officer should include specific instructions in requisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:27Z
Check the source to verify that this is the current version before use.

6.4 Subject term (key word) listing.

Airlift
Blivet
Collapsible
Drum
Elastomeric
Fabric
Fuel drum
Lifting
Lifting yoke
Tow yoke
Water drum
Yoke

6.4 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Air Force - 69

Review activities:
Army - GL
Air Force - 43 70 71 80, 99
DLA - GS

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 8110-0287

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:27Z
Check the source to verify that this is the current version before use.

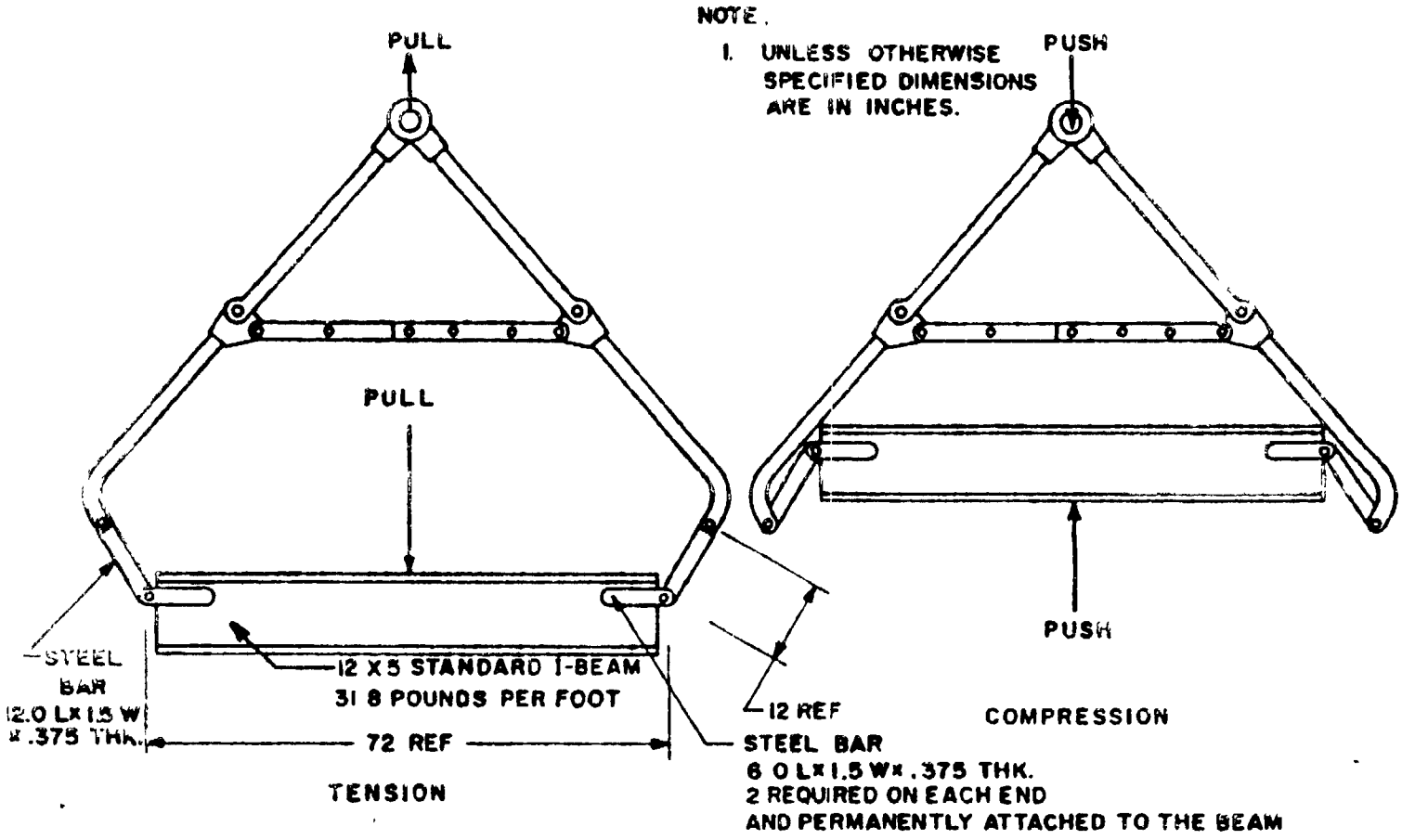


FIGURE I. Proof test.

X-166A

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:27Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Belvoir RDE Center
ATTN: STRBE-TSE
Ft. Belvoir, VA 22060-5606

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:27Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-Y-40628D	Yoke, Towing and Lifting: for Drums, Fabric, Collapsible

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITIONS ARE OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T19:27Z
Check the source to verify that this is the current version before use.